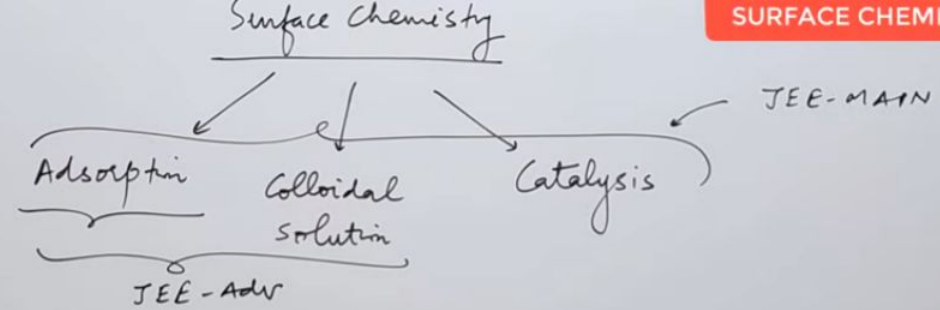

Surface Chemistry
JEE-MAIN
Adsorption
Colloidal solution
Catalysis
JEE-Adv

Adsorption : Phenomenon of accumulation or higher concentration of molecular species at the surface than in bulk is called adsorption.

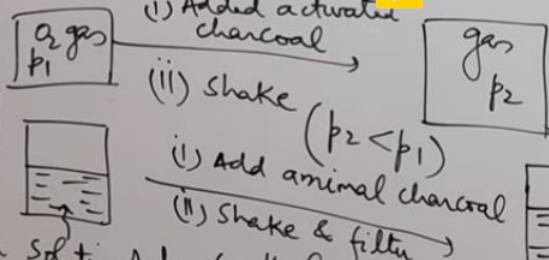
(A Surface phenomenon)

e.g. of physical adsorption

Adsorbate : Molecular species accumulated on surface.

Adsorbent : Surface provider (e.g → clay,

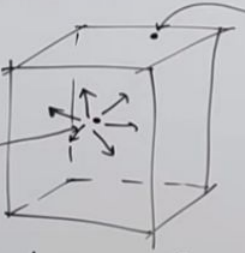
e.g of adsorption (I ϕ)



Common adsorbents
- → Powdered or Porous charcoal
- → Silica gel
- → alumina gel ← used in chromatography.
- → Powdered Pt/Rh etc.

SURFACE CHEMISTRY BY ALK SIR

Reason for adsorption: Reason for adsorption is the fact that surface has higher unbalanced residual forces than the bulk (infact in bulk of adsorbent, no residual forces as each molecule has similar environment.)



No residual force on molecule/atom in bulk.

charcoal block (adsorbent)

This surface atom/molecule of adsorbent has high surface enagy due to residual attractive forces so these molecules/atoms attract the adsorbate and results into accumulation of adsorbate.

If

SURFACE CHEMISTRY BY ALK SIR

Absorptin: e.g $H_2O_{(vap)} \xrightarrow[(Anhydrous)]{CaCl_2}$ water vapours absorbed by $CaCl_2$

(Absorptin)

e.g $H_2O_{(vap)} \xrightarrow[gel]{Silica}$ water vapours are adsorbed on surface of Silica gel.

(adsorptin)

Sorptin: When both absorptin & adsorptin takes places.

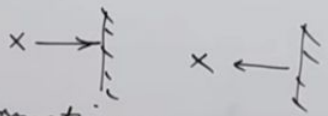
Desorptin: Reverse of adsorption.
(detachment of adsorbate from surface)

SURFACE CHEMISTRY BY ALK SIR

Type of adsorption : (Ip)

Physisorption (Physical Adsorption)

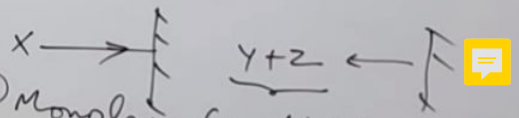
1. attachement through weak vanderwaal's forces.
2. Reversible



3. Multilayer formation occurs
4. Low enthalpy of adsorption.
   (small energy released, $\Delta H = -(20 \text{ to } 40)$ KJ/mol
5. Non-specific

chemisorption (chemical adsorption)

1. chemical bond (e.g covalent bond) formation between adsorbent & adsorbate.
2. Irreversible



3. Monolayer formation occurs.
   (layer after monolayer chemisorption, Physisorption may occur)
4. High enthalpy of adsorption
   $\Delta H = -(80 \text{ to } 240)$ KJ/mol
5. Highly specific

SURFACE CHEMISTRY BY ALK SIR

Type of adsorption : (Ip)

Physisorption (Physical Adsorption)	Chemisorption (chemical adsorption)
(6) Nil activation energy needed	(6) High activation energy.
(7) Occurs at low temperature	(7) Occurs at high temperature (generally) as energy is needed to cross high activation energy.
(4) Low enthalpy of adsorption. (Small energy released, $\Delta H = -(20 \text{ to } 40)$ KJ/mol	Physisorption may occur (4) High enthalpy of adsorption $\Delta H = -(80 \text{ to } 240)$ KJ/mol.
(5) Non-specific	(5) Highly specific

SURFACE CHEMISTRY BY ALK SIR

Thermodynamic parameters of adsorption:

→ For both Physisorption & chemisorption.

ΔH = -ve
ΔS = -ve
ΔG = -ve
} For adsorption.

$\Delta G = \Delta H - T\Delta S$

ΔH ↑ highly -ve.

$T\Delta S$ ↑ ΔS = -ve.

-ve is needed for adsorption to occur.

Factors affecting adsorption of gases over solid surface:

① **Nature of gas (adsorbate)**: Easily liquifiable gas ($H_2O_{(vap)}$, NH_3, SO_2 etc. which have high critical temp, $T_c = \frac{8a}{27Rb}$) are adsorbed to much greater extent than the permanent gases (e.g He, H_2, O_2, N_2 etc)

↑ In large range of temp, these permanent gases are gas only and liquify with great difficulty (have very low T_c)

② **Surface area of adsorbent**:

Surface area of adsorben ↑, extent of adsorption ↑.

⟶ extent of adsorption = $\frac{x}{m}$

where x = mass of gas adsorbed

m = mass of adsorbent.

⟶ Powdered form and porous form have high surface area.

Activated charcoal : Powdered charcoal.

② Surface area of adsorbent :
area

Surface of adsorben ↑, extent of adsorptin ↑.

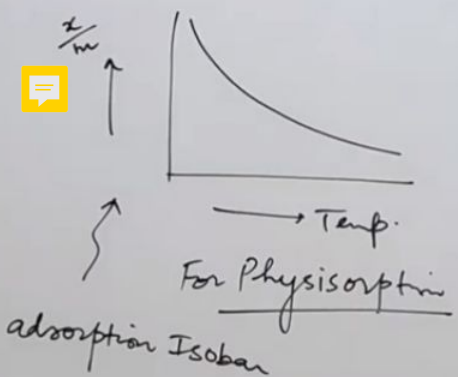
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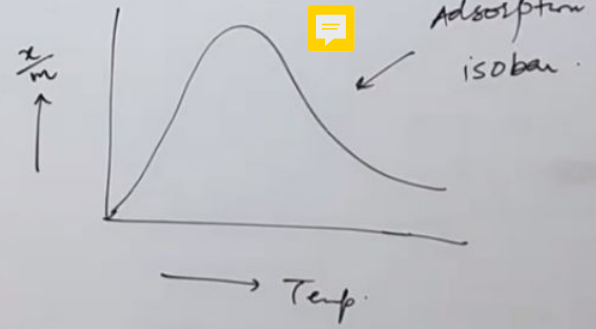
Activated charcoal : Powdered charcoal.

③ Temperature : (At Constant pressure of gas) ⇒ Isobars.



For Physisorption

adsorption Isobar



Chemisorption

→ In chemisorption, on increase in temp, extent of adsorption first increase (to overcome activation energy) and reaches max. and then decreases.

Q → which has/have correct increasing order of extent of adsorption on a solid surface?

(A) $CO_2 < O_2 < H_2$

(B) $H_2 < O_2 < CO_2$

(C) $He < N_2 < SO_2$

(D) $N_2 < H_2 < SO_2$

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(D) $N_2 < H_2 < SO_2$

e.g. $He < Ne < Ar < Kr < Xe$

increasing extent of adsorption

e.g. $H_2 < O_2 < CO_2$

increasing extent of adsorption.

④ Pressure: (At Constant temperature)

(Adsorption isotherms)

ⓐ Freundlich's adsorption isotherm:

⟶ Freundlich proposed an empirical formula for extent of adsorption

$$\frac{x}{m} = k p^{1/n}$$

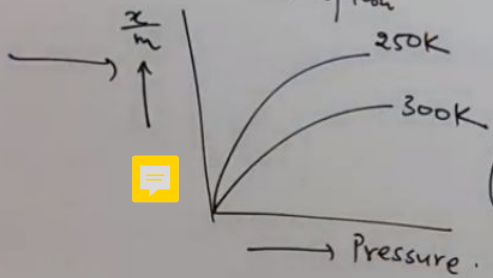
$p \longrightarrow$ pressure of gas,

$k, n \longrightarrow$ Constants depending upon the nature of adsorbent & adsorbate at given temp.

$\frac{1}{n}$ varies from 0 to 1

$[n > 1]$.

$\frac{x}{m}$ = extent of adsorption



(Physical adsorption).

SURFACE CHEMISTRY BY ALK SIR

Experimental determination of k, n

$$\frac{x}{m} = k p^{1/n}$$

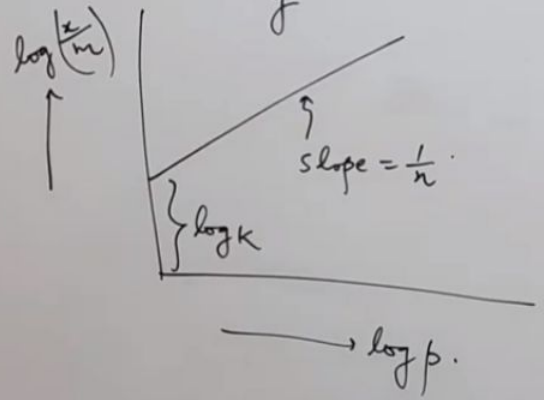
$$\log\left(\frac{x}{m}\right) = \log k + \frac{1}{n}\log p$$

y $\qquad$ x

$y = mx + c$

Slope $= \frac{1}{n}$

intercept $= \log k$



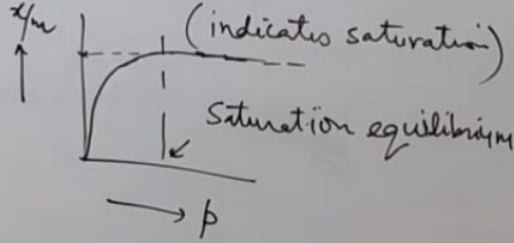
x/p

Note: The drawback of Freundlich adsorption isotherm is that it fails a high pressure and holds good over a limited range of pressure.

$$\frac{x}{m} = k p^{1/n}$$

If $\frac{1}{n} = 0$ (at high pr),

$$\frac{x}{m} = k \text{ (constant)}$$



SURFACE CHEMISTRY BY ALK SIR

Experimental determination of k, n

$$\frac{x}{m} = k p^{1/n}$$

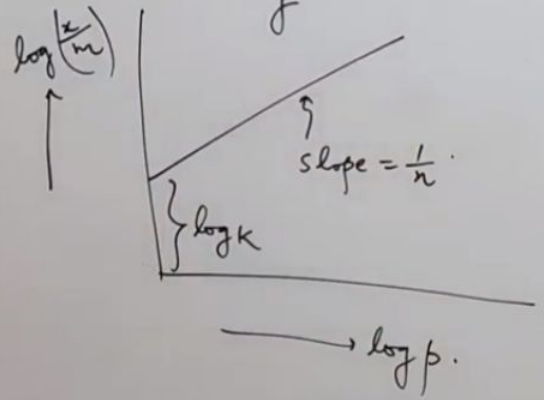
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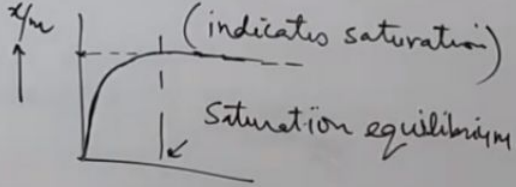


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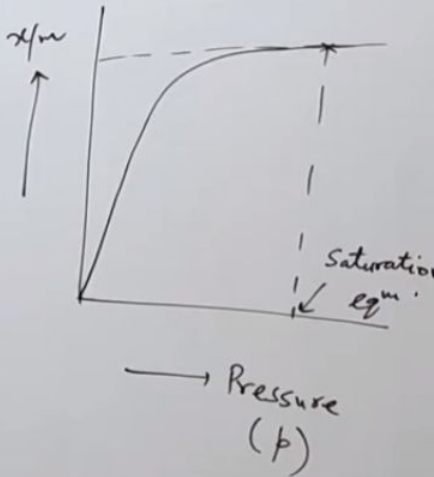


If $\frac{1}{n} = 1$ (at low pr.), $\frac{x}{m} = kp$

SURFACE CHEMISTRY BY ALK SIR

Langmuir adsorption isotherm : (Based on KTG)

$$\frac{x}{m} = \frac{ap}{1+bp}$$ where a & b are Langmuir's constants.



→ If pressure of gas is low, $bp << 1 \Rightarrow (1+pb) \cong 1$

$\therefore \frac{x}{m} = ap$ or $\frac{x}{m} \propto p$

→ At high pr, $bp >> 1 \Rightarrow (1+bp) \cong bp$.

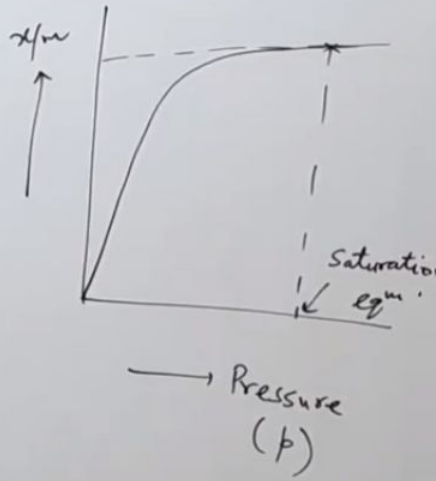
$$\frac{x}{m} = \frac{ap}{bp} = \frac{a}{b} = \text{Constt.}$$ (Saturation eqn).

→ There exists a dynamic eq^{m} at saturation point,

Rate of adsorption = Rate of desorption.

Imp

→ Since behaviour of gas molecules/atoms is ideal, the adsorption of gas molecule on surface site does not depend on whether the nearby site is vacant or occupied.



SURFACE CHEMISTRY BY ALK SIR

Note: → Freundlich & Langmuir adsorption isotherm (Physical adsorption) are valid for adsorption in solution phase. (Methylene blue dye adsorbed on animal charcoal)

Freundlich: $\frac{x}{m} = K C^{1/n}$ where C = Molar Conc. of solution.

Langmuir: $\frac{x}{m} = \frac{aC}{1+bC}$ where C = Molar Conc. of solution

Read NCERT

Applications of Adsorption:

1. To create vacuum: Activated charcoal.
2. Dehumidifiers: Silica gel is used.
3. In gas masks: Activated charcoal is used to adsorb poisonous gas like, CH_4, CO, SO_2 etc.

SURFACE CHEMISTRY BY ALK SIR

④ In chromatography : Alumina gel (Al_2O_3)

⑤ Froth floatation method : Pine oil adsorb sulphide ore and form froth.

⑥ Silica gel is used in Petroleum refining

⑦ In heterogeneous catalysis : Powdered Ni/Pt/Rh etc.
(In hydrogenation of alkene/alkyne)

Applications of Adsorption : ① To create vacuum : Activated charcoal.

② Dehumidifiers : Silica gel is used.

③ In gas masks : Activated charcoal is used to adsorb poisonous gas like, CH_4, CO, SO_2 etc.

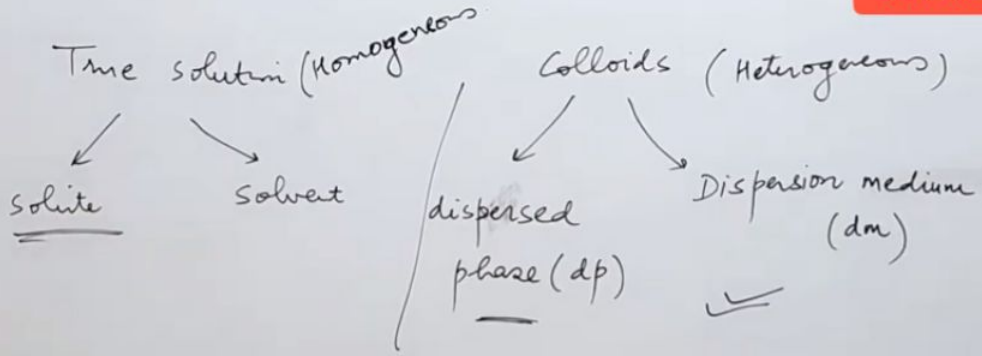
SURFACE CHEMISTRY BY ALK SIR

Colloids:

- homogeneous
  - True soln : salt or sugar solution.
- heterogeneous
  - Suspension : dust dissolved in water,
  - Colloids : butter, milk, paints etc.

Size of colloid is in between size of particles in true solution & suspension.

True solution	Colloids	Suspension
1 Å to 10 Å size	10 Å to 10,000 Å particle size	$> 10^4$ Å.
invisible	invisible	visible
Homogeneous	heterogeneous	Heterogeneous.
does not settle under gravity	Settles under Centrifuge	Settles under gravity
does not scatter light	scatters light	does not scatter light.


True solution (Homogeneous
Solute
Solvent
Colloids (Heterogeneous)
dispersed phase (dp)
Dispersion medium (dm)

SURFACE CHEMISTRY BY ALK SIR

Classification of Colloids :

(a) Based on physical state of dp & dm :

D.P.	D.M.	Type of colloid	e.g.
S	S	Solid sol	gemstone, coloured glass.
S	L	sol	gold sol, Pt sol, S sol., As_2S_3 sol etc.
S	G	aerosol	Smoke, dust
L	S	gel	cheese, jellies
L	L	emulsion	milk, butter, hair cream
L	G	liq. aerosol	mist, fog, cloud.
G	S	solid foam	Pumice stone, foam rubber.
G	L	foam	soap lather, beer froth.

SURFACE CHEMISTRY BY ALK SIR

(b) based on interaction between dp and dm :

(i) Lyophilic sol : Strong affinity between dp & dm,
(Solvent loving) e.g glue, gelatin, protein, starch etc.

(Ip)

(ii) Lyophobic : weak interaction between dp & dm.
(Solvent hating) e.g Au sol, Ag sol, 'S' sol, As_2S_3 sol etc.

(12)

Lyophilic sol

1. prepared by just mixing and shaking i.e. easily prepared.
2. Reversible
3. Highly stable
4. Surface tension of sol < surface tension of d.m.
5. Viscosity of sol > viscosity of d.m.
6. Generally electrolyte are not able to cause coagulation (or precipitation)

Lyophobic sol

1. special methods are used to prepare lyophobic sols.
2. Irreversible.
3. Unstable, infact requires stabilizer.
   (e.g emulsions (e.g milk) is unstable and to stabilize it 'Casien' is added).
4. Surface tension of sol ≡ Surface tension of d.m.
5. Viscosity of sol ≡ viscosity of d.m.
6. Easily coagulated by electrolytes.

SURFACE CHEMISTRY BY ALK SIR

(C) based on charge on colloid particles :

V.Imp

(i) +vely charged sol : → Imp → metal oxide & metal hydroxide sol.

e.g SnO_2, TiO_2, Al_2O_3 sol etc.

e.g $Cr(OH)_3$, $Al(OH)_3$, $Fe(OH)_3$ etc.

→ Basic dyes e.g methylene blue.

(ii) −vely charged sol :

→ metal sols.

Au sol, Ag sol, Pt sol.

→ Sulphide sols.

e.g CdS sol, As_2S_3 sol, Sb_2S_3 sol etc.

Imp

* Haemoglobin present in blood → Blood, clay etc.

is +ve sol but blood overall is −ve sol.

SURFACE CHEMISTRY BY ALK SIR

(D) based on size of colloid particle (or d/p particle) → Multimolecular, Macromolecular, Associated colloid (Miscelles)

(i) Multimolecular colloid: → Small atoms/Molecule associate to form particles of colloidal size (1 nm to 1000 nm)

→ particles (atoms or molecules) associate through vanderwaal's forces.

→ e.g 'S' sol, 'Au' sol, 'Pt' sol, 'Ag' sol etc

(ii) Macromolecular colloid: Single particle (molecule) is large enough in size to be in colloidal size range.

e.g protein, cellulose, starch, haemoglobin.

(iii) **Associated Colloid (miscelles)** : These are usually the solution of sod. or pot. salt of fatty acids which act as strong electrolyte true solution at low concentration but act as colloid when concentration is high.

Saturated fatty acid salts.

e.g { Sod. palmitate, $C_{15}H_{31}COO^- Na^+$ } Soaps
{ Sod. stearate, $C_{17}H_{35}COO^- Na^+$ }

e.g Sod. lauryl sulphate, $CH_3(CH_2)_{11}SO_4^- Na^+$ } (detergent)

(iii) Associated Colloid (miscelles): These are usually the solution of sod. or pot. salt of fatty acids which act as strong electrolyte true solution at low concentration but act as colloid when concentration is high.

Saturated fatty acid salts.

e.g { Sod. palmitate, $C_{15}H_{31}COO^- Na^+$; Sod. stearate, $C_{17}H_{35}COO^- Na^+$ } Soaps

e.g Sod. lauryl sulphate, $CH_3(CH_2)_{11}SO_4^- Na^+$ (detergent)

Kraft temperature (T_k): The minimum temperature below which miscelle formation does not occur.

Critical miscelle Concentration (CMC): Minimum Conc. above which miscelle formation can occur.

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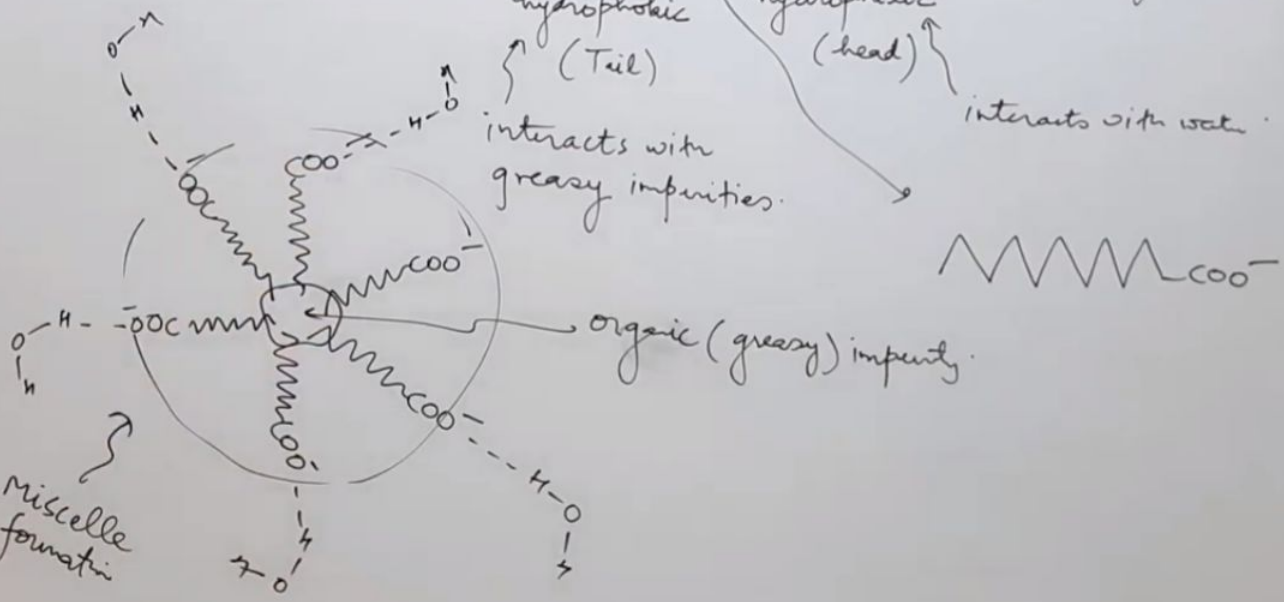
Info → Longer is the hydrophobic chain (alkyl part), smaller will CMC value.

Mechanism of miscelle formation :

$C_{17}H_{35}COO^-Na^+$ (soap). (acts as surfactant)

hydrophobic (Tail) — interacts with greasy impurities.

hydrophilic (head) — interacts with water.



→ Cleansing action of soap is due to miscelle formation. Infact soap or detergent are called surfactants which reduce interfacial tension between dp (organic greasy impurity) and dm (water).

V.V.Sp

V.Sp

→ Thus soap/detergent help in emulsification and washing away of oil and fats.

Lp

* Soap/detergent are emulsifiers.

→ * Caesien (a protein molecule) is emulsifier in milk which binds both liquid fat (dp) and water (dm) in milk.

Lp

[Emulsions are unstable colloid and need a stabilizer like Caesien for milk)

SURFACE CHEMISTRY BY ALK SIR

Preparation of colloids: (For Lyophobic sols)

(A) Condensation method:

(1) Chemical method (e.g double decomposition / Redox / ppt^n etc).

→ As_2O_3 + H_2S (aq) → As_2S_3 sol + H_2O
(in hot water) (saturated solution) ↑ double decomposition Rex.

→ SO_2 + H_2S (aq) → S (sol) + H_2O
(OA) (Saturated sol^n)

→ $2AuCl_3$ + $3HCHO$ + $3H_2O$ → $2Au$ (sol) + $3HCOOH$ + $6HCl$
(O.A) (R.A) (Purple of Cassius)

→ $AgNO_3$ + tannic acid / NH_2NH_2 → Ag sol

→ $FeCl_3$ (aq) + H_2O (l) → $Fe(OH)_3$ sol + HCl.

(ii) Exchange of solvent: when a true solution is mixed with a solvent in which solute of the true solution is insoluble but both the solvents are miscible, then colloid is formed.

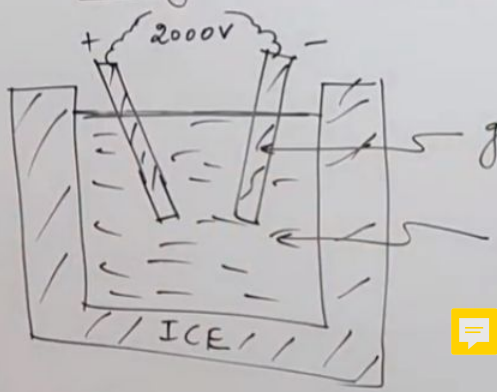
e.g. 'S' in alcohol + $H_2O(l)$ ⟶ 'S'(sol).
(true solution) (excess)

(iii) Excessive cooling: Colloidal sol of ice can be prepared by cooling a mixture of water & $CHCl_3$.

Dispersion method :

(i) **Mechanical method** : A ball mill is used.
(used to prepare printing ink)

V.I.P (ii) **Bredig's electric arc method** :



Preparation of gold sol

I.P → Involves both dispersion and condensation

I.P → Sol of less active metals like Au, Pt, Cu, Ag etc are prepared by this method.

→ Intense heat is generated by high P.D which vaporise the metal. metal atoms in cold dm climb to form metal sol.

(iii) Peptization : Pepsin (an enzyme) which breaks protein molecule of food and helps in digestion of food.

When freshly prepared (precipitated) compound is mixed with solvent and a small amount of electrolyte solution is added in it, then a colloidal solution is formed. Electrolyte is called peptizing agent.

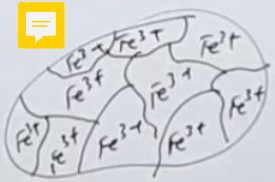
e.g Preparation of $Fe(OH)_3$ sol.

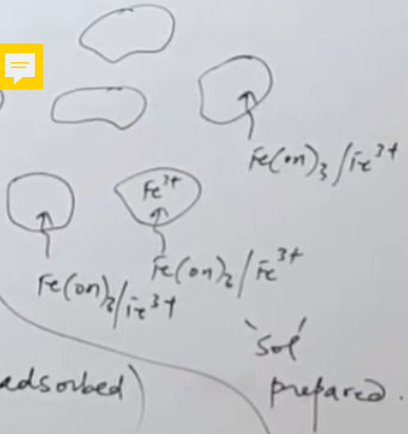
$$\underset{\text{(fresh ppt)}}{Fe(OH)_{3(s)}} + \underset{\text{Electrolyte (Peptizing agent)}}{FeCl_{3(aq)}} \longrightarrow \underset{\text{(+ve ferric hydroxide sol)}}{Fe(OH)_3/Fe^{3+}\ sol}$$



+ $FeCl_{3(aq)}$

(electrolyte)





$Fe(OH)_{3(s)}$

ppt

Fe^{3+} gets adsorbed on $Fe(OH)_3$ crystals.

(Common ion is easily adsorbed)

$Fe(OH)_{3(s)}$ + $FeCl_{3(aq)}$ ⟶ $Fe(OH)_3/Fe^{3+}$ sol

(fresh ppt)

↑

Electrolyte (Peptizing agent)

($Fe(OH)_3/Fe^{3+}$ sol)

(+ve ferric hydroxide sol)

SURFACE CHEMISTRY BY ALK SIR

Purification of colloid : It is done by

(i) Dialysis (removal of impurities from sol by semipermeable membrane (spm) is called dialysis)

(ii) **Ultrafiltration** : Very fine filters are used.

(iii) **Ultracentrifugation** : (Under centrifuge at high rpm (rev. per min), colloid particles settle down)

SURFACE CHEMISTRY BY ALK SIR

Properties of Colloidal sol :

① **Colligative property** : → Colloidal sol show colligative property of low order.

Imp → Measurement of osmotic pressure is most accurate to determine molar mass of colloid particles.

② **Optical property** : → When a beam of light is passed through a colloid sol kept in dark room, the colloidal particles become visible due to scattering of light. The phenomenon is called Tyndall effect and the beam is called Tyndall cone.

→ Tyndall effect is very weak for lyophilic colloids.

e.g. of Tyndall effect: ① Tail of comets (visibility ↑)

② Blue colour of sky & sea water.

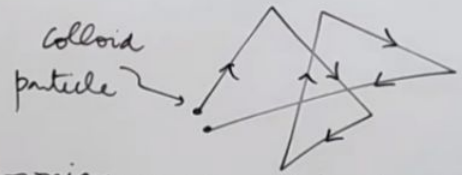
② optical property: → when a beam of light is passed through a colloid sol kept in dark room, the colloidal particles become visible due to scattering of light. The phenomenon is called Tyndall effect and the beam is called Tyndall cone.

→ Tyndall effect is very weak for lyophilic colloids.

e.g. of Tyndall effect: ① Visibility Tail of comets

② Blue colour of sky & sea water.

③ Mechanical Property: Colloidal particles are in a state of continuous Random motion following zig-zag path, called Brownian Movement.



⟶ This Brownian movement is due to collision of colloid particle with dm particles.

Tyndall effect is observed only when these two conditions are satisfied:

(Sp) (i) diameter of colloid particle (dp particle) should not be much smaller than the wavelength of light used.

(Sp) (ii) Refractive indices of dp and dm should have large difference.

JEE

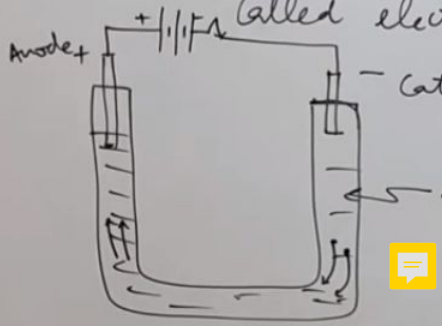
(4) Electrical properties :

→ Sol particle (dp particles) are either +ve or −ve charged.

e.g metal sols & metal sulphide sols : −ve charged.

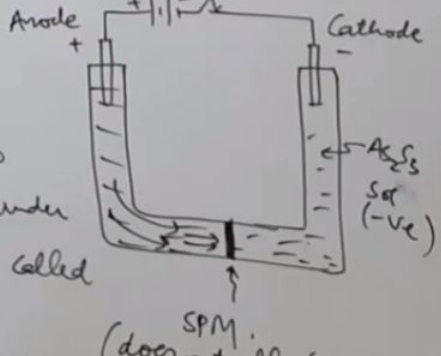
e.g metal oxide & metal hydroxide sol : +ve charged.

Electrophoresis : Movement of charged colloid particles (dp particle) under potential difference towards respective electrode is called electrophoresis.



Electroosmosis :

→ movement of dm particles towards respective electrodes under P.D through SPM is called electroosmosis.



Electrophoretic mobility : It is the speed of colloid particles (dp particles) under unit potential gradient.

Isoelectric point : The pH of solution at which colloid particles (or dp particles) neither move towards cathode nor towards anode under applied potential difference as the dp particls become unchanged, is called isoelectric point.

⟶ At isoelectric point, sol partides become unchaged and starts coagulation (or precipitation)
(Sol is unstable at isoelectric point)

Stability of sol : Stability of solution is attributed mainly to the following.

(i) Brownian movement

(ii) Similar charge present on colloid (dp) particles.

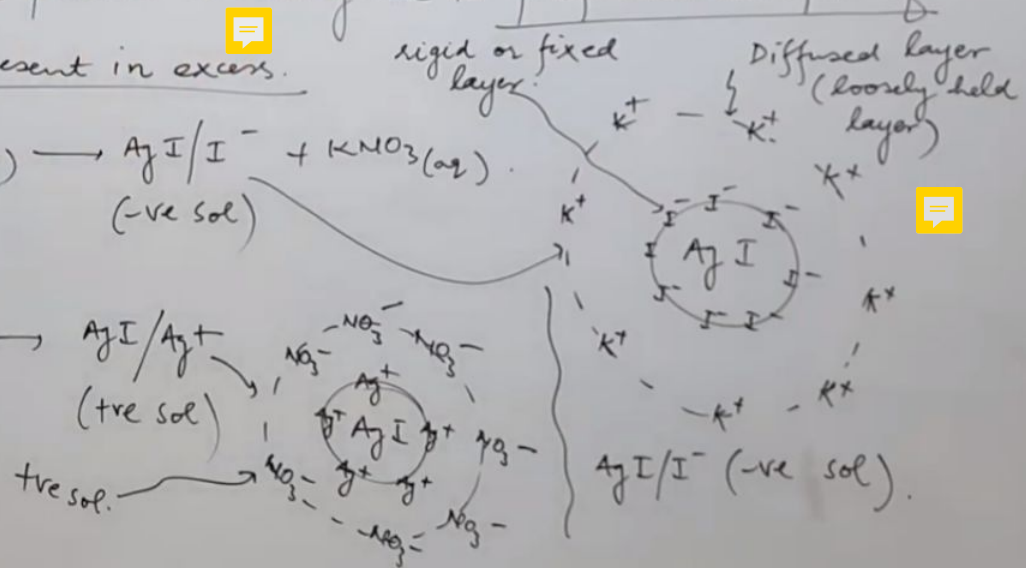
(V.Imp)

→ charge on colloid particles is mainly due to preferential adsorption of common ion present in excess.

→ e.g $KI_{(aq)} + AgNO_{3(aq)} \longrightarrow AgI/I^- + KNO_{3(aq)}$
(excess) (L.R) (-ve sol)

→ e.g $KI_{(aq)} + AgNO_{3(aq)} \longrightarrow AgI/Ag^+$
(excess) (+ve sol)

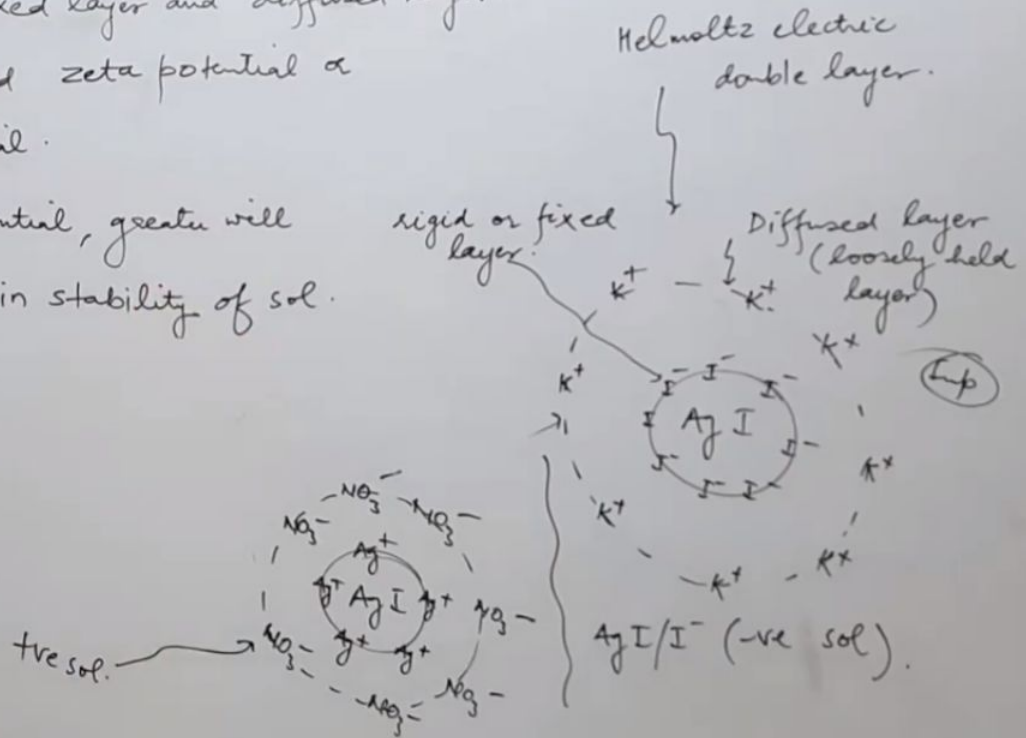
+ve sol. →



(xx) → In Helmoltz electric double layer, there exists a potential difference between fixed layer and diffused layer.

This P.D is called zeta potential or electrokinetic potential.

(Imp) → Higher is zeta potential, greater will be its contribution in stability of sol.

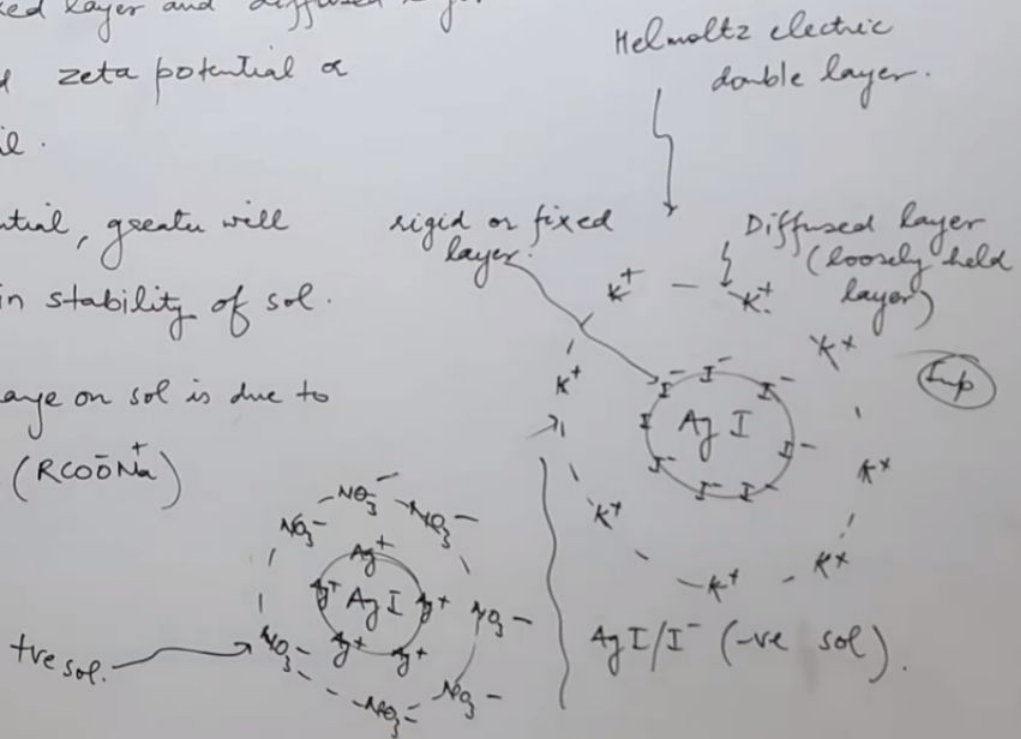


→ In Helmoltz electric double layer, there exists a potential difference between fixed layer and diffused layer.

This P.D is called zeta potential or electrokinetic potential.

(Imp) → Higher is zeta potential, greater will be its contribution in stability of sol.

→ In soap/detergent, charge on sol is due to ionisation of electrolyte ($RCOO^-Na^+$)



Coagulation of sol :

→ When electrolyte is added in large conc. to sol solution, the sol particles take up the ion of electrolyte which is opposite in charge and thus gets neutral. Neutral sol particles start aggregating (through vanda waal's forces of attraction) and precipitate (or coagulate).

→ Note that a small amount of electrolyte is needed for stability of sol. But adding electrolyte in large conc. causes coagulation.

→ Flocculation : when associated colloid particles (or coagulated sol particles) can be broken into particles of colloidal size just by shaking

Coagulation : when associated colloid particles can't be broken into particles of colloid size by just shaking.

→ Flocculation ; when associated colloid particles (or coagulated sol particles) can be broken into particles of colloidal size just by shaking

Coagulation : when associated colloid particles can't be broken into particles of colloid size by just shaking.

V.Imp

→ Coagulation of lyophobic sol can be done by

(i) Electrophoresis

(ii) Mutual precipitation or coagulation (when a +ve & -ve sol are mixed)

(iii) By addition of electrolyte

(iv) By Boiling.

V.IMP

Hardy Schulze rule :

→ Coagulation power of an electrolyte is linked with the valency of ion causing coagulation of sol.

→ higher is valency of ion of electrolyte causing coagulation, greater will be the coagulation power of electrolyte.

→ To coagulate -ve sol,

$Al^{3+} > Ba^{2+} > K^{+}$

decreasing value of coagulation power.

Q→ To Coagulate As_2S_3 sol, which is most effective?

(A) $K_4[Fe(CN)_6]$, (B) $AlCl_3$

(C) $Ba(NO_3)_2$ (D) Mg_3PO_4.

Soln : As_2S_3 (-ve sol)

SURFACE CHEMISTRY BY ALK SIR

Coagulation value: It is the amount of electrolyte in 'mmol' that should be added in 1 lit of colloidal solution to completely coagulate it.

Coagulation power $\propto \frac{1}{\text{Coagulation value}}$.

Protective Colloid: → Lyophilic colloids are much more stable than the Lyophobic colloids and hence lyophilic colloid act as protective colloid.

→ When a lyophilic colloid is added as protective colloid into lyophobic colloid, a protective layer of lyophilic colloid is formed outside the lyophobic colloid. Now electrolyte is not able to coagulate lyophobic colloid.

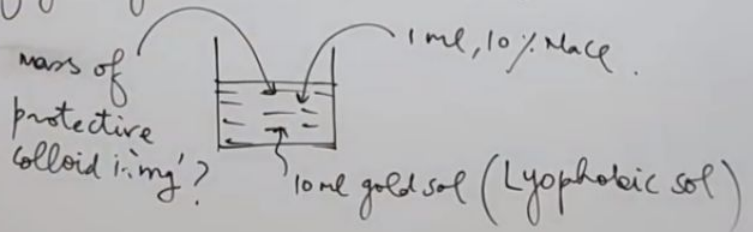
SURFACE CHEMISTRY BY ALK SIR

→ Gelatin protects gold sol (lyophobic colloid) from coagulation on the addition of NaCl solution (electrolyte)

a very effective protective colloid.

→ Protective power of lyophilic colloid is measured in terms of Gold Number.

Gold number (For protective Colloid): It is defined as the minimum amount of protective colloid (in mg) required to just prevent the coagulation of 10 ml of given gold sol when 1 ml, 10% NaCl solution is added to it.



⟶ Lower is the gold number, greater will be its protective power.

	Gold No.
✓ Gelatin	0.005 – 0.01
Haemoglobin	0.03 – 0.07
Egg albumin	0.1 – 0.2
Gum arabic	0.15 – 0.25
Starch	20 – 25

Note: (IP) Gelatin is added in preparation of icecream to protect the particles of ice.

SURFACE CHEMISTRY BY ALK SIR

Emulsions : → dp → Liquid

dm → Liquid

→ They are liquid in liquid immiscible or partially miscible components.

→ Unstable and requires stabilizer.

→ Type → Oil in water type (O/W) : e.g milk,
→ To stabilize milk, protein 'Casein' is present as emulsifier.
Vanishing cream. (PONDS/NIVIA)

→ Water in oil (W/O) type.
e.g. butter, hair cream.

⟶ Components of emulsion can be separated by

(i) Centrifugation or

(ii) Freezing or

(iii) Boiling.

Colloids around us & applications:

① **Artificial rain**: ⟶ Electrified sand or AgI is sprayed over clouds by aeroplane to cause coagulation of cloud (liquid water(dp) in air(dm)).

② **Formation of Delta**: ⟶ River water contain colloids and when meet with sea water (contain electrolytes, e.g $CaCl_2$, $MgCl_2$, NaCl etc), Coagulation occurs and solid deposits accumulate, called Delta formation.

SURFACE CHEMISTRY BY ALK SIR

(3) Cottrell smoke precipitator : → Exhaust (smoke) of factory/industry are let off through chimney and this smoke causes air pollution.

→ In chimney, cottrell smoke precipitator is fitted in which a high P.D. Causes charged smoke to get precipitated as ash (राख)

(4) In medicine :

→ Alum is used to purify drinking water

→ Colloidal gol is used for intramuscular injection (Inj)

→ Colloidal antimony is used to cure Kalaazar (काला ज्वर, Black fever)

→ Milk of magnesia is used to treat stomach disorders.

SURFACE CHEMISTRY BY ALK SIR

Catalysis :

Catalyst : which changes or alter the rate of a chemical reaction without change in its structure chemically and without being consumed.

e.g $KClO_{3(s)} \xrightarrow[\Delta]{MnO_2} KCl_{(s)} + 1.5O_{2(g)}$.

(MnO_2 : Catalyst)

Promotor : Substance which enhance the activity of Catalyst

e.g $N_{2(g)} + 3H_{2(g)} \xrightarrow[+Mo_{(s)}]{Fe_{(s)}} 2NH_{3(g)}$ [Haber's process]

(Fe : Catalyst, Mo : Promotor)

Poison : which decrease the activity of Catalyst

e.g Rosenmund's reduction (V.Imp)

$RCOCl + H_2 \xrightarrow[BaSO_4 (Poison)]{Pd (Catalyst)} RCHO + HCl$.

SURFACE CHEMISTRY BY ALK SIR

Homogeneous catalyst: when catalyst and reactants are in same phase.

e.g. $2SO_{2(g)} + O_{2(g)} \xrightarrow{NO_{(g)}} 2SO_3$ (NO: Catalyst)

e.g. $CH_3COOCH_{3(l)} + H_2O_{(l)} \xrightarrow{HCl_{(aq)}} CH_3COOH_{(aq)} + CH_3OH_{(aq)}$. (HCl: Catalyst)

Heterogeneous catalyst: when reactants & catalyst are not in same phase

e.g. $N_{2(g)} + 3H_{2(g)} \xrightarrow{Fe_{(s)}} 2NH_{3(g)}$. (Haber's process)

e.g. $4NH_{3(g)} + 5O_{2(g)} \xrightarrow{Pt_{(s)}} 4NO_{(g)} + 6H_2O_{(g)}$. (Ostwald's process)

SURFACE CHEMISTRY BY ALK SIR

(V.I.P)

Adsorption theory of Heterogeneous catalysis :

Mechanism involves the following 5 steps

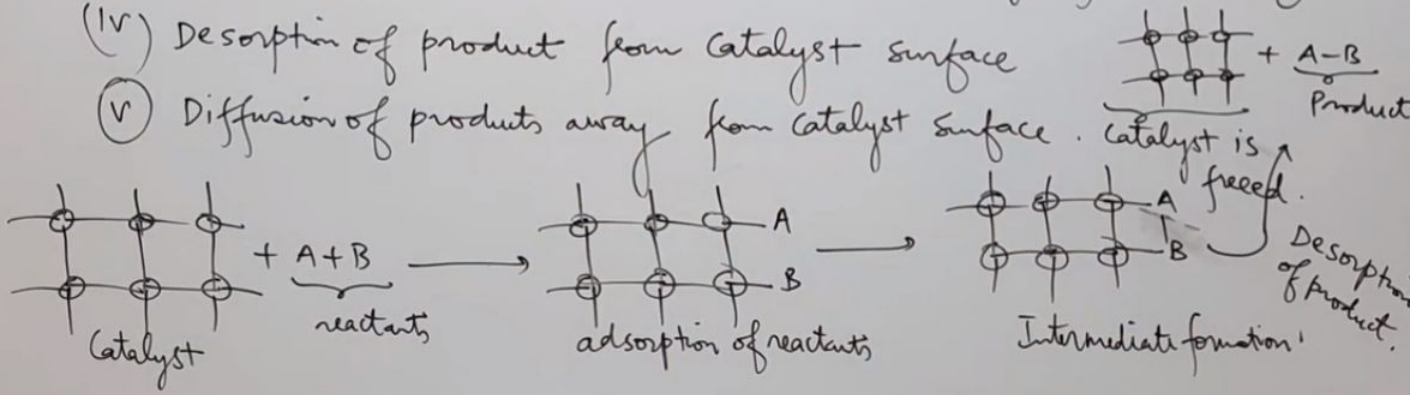
(i) Diffusion of reactants on surface of catalyst

(ii) Adsorption of reactants on surface sites of catalyst (chemical adsorption)

(iii) chemical reaction between reactants through formation of intermediate

(iv) Desorption of product from catalyst surface

(v) Diffusion of products away from catalyst surface. Catalyst is freed.



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Important features of solid catalyst :

① **Activity :** (1p) → It represents strength of chemisorption of reactants on catalyst surface.

(1p) → Both the reactants should have nearly similar activity on catalyst surface. If one reactant has very high activity, then it will adsorb to all the surface sites leaving no place for other reactants to get adsorbed.

(1p) → Catalytic activity increases from group 5 to 11 metals with maximum activity shown by group 7 to 9.

e.g $2H_{2(g)} + O_{2(g)} \xrightarrow{Pt} 2H_2O_{(l)}$

SURFACE CHEMISTRY BY ALK SIR

② Selectivity: It is the ability of catalyst to direct a reaction to yield a particular product.

e.g $CO_{(g)} + 3H_{2(g)} \xrightarrow{Ni} CH_{4(g)} + H_2O_{(g)}$

$CO_{(g)} + 2H_{2(g)} \xrightarrow{Cu/ZnO-Cr_2O_3} CH_3OH_{(g)}$

$CO_{(g)} + H_{2(g)} \xrightarrow{Cu} HCHO_{(g)}$.

Shape Selective Catalysis by Zeolites :

→ Zeolites have honeycomb structure, having microporous 3D-aluminosilicates (Si-O-Al) framework.

→ It depends on size and shape of reactants and products and size of pores/cavities present on zeolites.

→ They are natural (found in nature) as well as synthesised in lab. for catalytic selectivity

Visit

→ ZSM-5 (a zeolite catalyst) is used in petroleum industry. It converts alcohol directly into gasoline (Petrol)

SURFACE CHEMISTRY BY ALK SIR

Enzyme Catalysis :

→ Enzymes are complex nitrogeneous organic compounds which are produced by living plants and animals.

→ They are protein molecules of large molecular mass which form colloidal solution in water.

→ e.g. Inversion of Cane sugar

$$\underset{\text{(Sucrose)}}{C_{12}H_{22}O_{11}(aq)} + H_2O(l) \xrightarrow[\text{(enzyme catalyst)}]{\text{Invertase}} \underset{\text{glucose}}{C_6H_{12}O_6} + \underset{\text{fructose}}{C_6H_{12}O_6}$$

e.g. $\underset{(C_6H_{12}O_6)}{\text{glucose}} \xrightarrow{\text{Zymase}} 2C_2H_5OH + 2CO_2$

e.g. $2(C_6H_{10}O_5)_n + nH_2O(l) \xrightarrow{\text{Diastase}} n\,C_{12}H_{22}O_{11}$

Starch → Maltose

SURFACE CHEMISTRY BY ALK SIR

e.g. $C_{12}H_{22}O_{11} + H_2O_{(l)} \xrightarrow[\text{enzyme}]{\text{Maltase}} 2C_6H_{12}O_6$
(Maltose) glucose

e.g $NH_2CONH_2 + H_2O_{(l)} \xrightarrow{\text{Urease}} 2NH_3 + CO_2$

e.g Enzyme 'pepsin' converts protein into peptides in stomach.

e.g 'Lactobacilli' enzyme converts milk into curd.

→ e.g Inversion of Cane sugar

$C_{12}H_{22}O_{11(aq)} + H_2O_{(l)} \xrightarrow[\text{(enzyme catalyst)}]{\text{Invertase}} C_6H_{12}O_6 + C_6H_{12}O_6$
(Sucrose) glucose fructose.

e.g glucose $(C_6H_{12}O_6) \xrightarrow{\text{Zymase}} 2C_2H_5OH + 2CO_2$

e.g $2(C_6H_{10}O_5)_n + nH_2O_{(l)} \xrightarrow{\text{Diastase}} n\,C_{12}H_{22}O_{11}$
Starch Maltose

SURFACE CHEMISTRY BY ALK SIR

characteristics of enzyme catalyst :

① **Very highly efficient** : Even a single enzyme molecule can convert a million of reactant molecules into products.

② **Highly specific**.

③ **Highly active under optimum temperature**.

→ optimum temp. range is 298 K to 310 K

↑ or 37°C temp (human body temp.)

④ **Highly active under optimum pH** :

→ Most active when pH is 5 to 7.

⑤ **Activators** and **Co-enzymes**

↑ eg metal ions like Na^+, Mn^{2+}, Co^{2+}, Cu^{2+}

↑ e.g non-protein molecules like vitamins.

⑥ Poison: which decrease activity of enzyme.

→ Poisons or inhibitors interact with active function groups (−NH₂, −OH, −SH etc) of enzyme on enzyme surface and destroy the catalytic activity of enzyme.

④ Highly active under optimum pH:

→ Most active when pH is 5 to 7.

⑤ Activators and Co-enzymes

Activators: eg metal ions like Na^+, Mn^{2+}, Co^{2+}, Cu^{2+}

Co-enzymes: e.g non-protein molecules like vitamins.

SURFACE CHEMISTRY BY ALK SIR

Mechanism of enzyme Catalysis : (Key & Lock Mechanism)

→ There are a number of cavities of characteristic shape present on surface of colloidal enzyme.

→ These cavities possess active groups such as $-NH_2$, $-COOH$, $-OH$, $-SH$ etc.

→ The molecule of reactant (or substrate) which has complimentary shape, fits into these cavities just like key fits into lock.

